

fields and chemical reactions far better than conventional storage media.

However, encoding data traditionally requires multiple laser pulses for each voxel, making the process relatively slow. To overcome this limitation, researchers from Microsoft's Silica project developed two different types of voxels: phase voxels and birefringent voxels.

In the first case, data is stored by changing the refractive index through isotropic changes in the glass. In the second, the voxel is polarised through anisotropic changes. Both types of voxels can be created using a single laser pulse, significantly improving efficiency. Using this method, researchers can create up to ten million voxels per second, allowing 25.6 million bits of data to be encoded every second.

Kazansky and his colleagues developed the underlying physics behind laser-writing technology and thus achieved the Guinness World Record for the most durable digital preservation using fused silica glass. In 2017, Microsoft began developing its Project Silica technology based on its work. Although fused silica is more durable and capable of storing data at higher densities, recent research has shown that Microsoft instead uses relatively inexpensive borosilicate glass, which is easier to manufacture. Researchers have also found experimentally that using four laser beams in parallel significantly speeds up the writing process.

At present, pen drives and hard drives remain the most common tools for everyday data transfer. Most modern storage devices rely on magnetic technology, storing information by creating different magnetic patterns on thin metal layers inside magnetic tapes and hard drives. However, as temperatures rise over time, the magnetisation of these tiny magnetic regions gradually weakens, making long-term data storage less reliable.

By contrast, once information is stored in glass, it is extremely difficult to erase and requires little ongoing maintenance or temperature control. Scientists have calculated that even at temperatures of about 290°C, the stored data remains unaltered. As a result, once data is written onto the glass, it can be preserved for an exceptionally long time.

Mark Bathe, a biotechnologist at the Massachusetts Institute of Technology, described this glass-based data storage system as an excellent solution for long-term archival storage. Many experts believe the technology could become an important option for preserving valuable digital records far into the future.

However, the biggest limitation is that the glass-based system is non-rewritable and unsuitable for storing frequently accessed data, making it impractical for everyday storage or data transfer. Even so, researchers remain optimistic that continued advances will expand its practical applications in the years ahead. **EFY**

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